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Yale University student studying Computer Science and Mathematics with experience in machine learning, high-performance computing, and production software engineering. Interested in AI, distributed systems, and performance-critical software.

Education

Yale University – New Haven, CT

- Graduation date: May 2028, B.S. in Computer Science and Mathematics (Intended)
- Relevant Coursework:
 - Computer Science: Data Structures and Algorithms, Systems Programming, Proof-based Algorithm design
 - Math and Statistics: Proof-based Linear Algebra, Advanced Applied Linear Algebra, Real Analysis, Probability Theory, Theory of Statistics

Professional Experience

Software Development Intern, Interactive Brokers

June 2026 – Present

- Designed and implemented digital twin infrastructure validation using Containerlab to recreate production network topologies in a CI/CD pipeline, automating pre-deployment testing of infrastructure changes and reducing deployment risk.
- Developed a statistical anomaly detection system using Prometheus/Thanos, STL (LOESS), and z-score analysis to identify abnormal disk utilization and network traffic across production exchange servers.
- Achieved the highest portfolio return among 8 teams in a month-long simulated trading competition.

Machine Learning Research Assistant, Economics Department, Yale University

May 2025 - Present

- Designed and implemented software solutions for a tunable ML model generating a satellite-imaging-based wealth index using geospatial data.
- Collaborated with faculty and graduate researchers to develop ML models and generate raster and statistical visualizations.
- Leveraged Yale's High-Performance Computing Cluster for large-scale batch jobs and GPU-accelerated model training/testing.
- Applied advanced ML engineering techniques, including parameter optimization, tensor manipulation and encoder/decoder models to improve model accuracy and scalability.

Skills and Projects

Personal Projects

- **Shakespearean Chatbot:** Designed and trained a transformer-based chatbot on the complete works of Shakespeare to generate responses in Elizabethan style.
- **Coinbase Trading Bot:** Built an automated trading system leveraging Coinbase's API for high-speed transaction monitoring, integrating LSTM models for price prediction and strategy optimization.

Skills

- Python, C, JavaScript, PyTorch, TensorFlow, NumPy, Pandas, React, MongoDB, Node.js, Git, High-Performance Computing, GPU utilization

Additional Information

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- **Languages:** English-Native, Spanish-Advanced Level
 - **For Fun:**
 - Violist, State-ranked player in High school, Yale Symphony Orchestra (2024-2025)
 - Chess player, 1650 online rating, Recreational Tennis and Ping Pong